Does Electrocution Improve Roller-Crimper Cover Crop Termination?

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Poster board 1251

Introduction

- Organic growers cannot use herbicides, so cover crops must be killed mechanically; incomplete kill causes regrowth, weed pressure and planting problems **(Creamer and Dabney 2002)**.
- The roller crimper (RC) kills cereal rye reliably only after anthesis, and surviving tillers regrow **(Mirsky et al. 2009; Kornecki 2015)**.
- Electrical weed control (EWC) kills plants with high-voltage current **(Slaven et al. 2023)**, but one pass only top-killed rye **(Sullivan 2025)**.
- **Gap:** whether EWC can kill what the crimper leaves alive, and whether pass order matters, is unknown.

Objective and Hypothesis

Objective: Determine whether EWC, before or after roller crimping, improves cereal rye termination over either tool alone.

Hypothesis: EWC + RC terminates rye better than either tool alone, in either order: crimping crushes the stems, and current kills the tillers the crimper misses.

Materials and Methods

Experimental design

- College Station, TX, 2026; RCBD, 4 reps
- Cereal rye (*Secale cereale* L.), Zadoks 85
- Sequences: same-day passes (Fig. 1)

5 TREATMENTS
Control · EWC · RC
RC→EWC · EWC→RC

COVER CROP
Cereal rye, soft dough
≈ 615 g m^{-2} dry biomass

EWC: ZASSO EH60
2.5 m, 3.2 km h^{-1}
One pass

ROLLER CRIMPER
Commercial, 3 m wide
One pass

Measurements

- Termination and survival: 3, 7, 15, 28 DAT
- Biomass (0, 28 DAT); soil moisture

EWC energy dose

- P = 3 × 255 V × 40–45 A ≈ 31–34 kW
- **E/A = P / (v · w) ≈ 13.8–15.5 kJ m^{-2}**

Data analysis

- RCBD ANOVA; Tukey's HSD (α = 0.05)
- Python 3.11, statsmodels 0.15 **(Seabold and Perktold 2010)**



Fig 1. EWC (Zasso EH60) and roller crimper in cereal rye.

Results and Discussion



Fig 2. Plots 15 days after treatment (DAT).

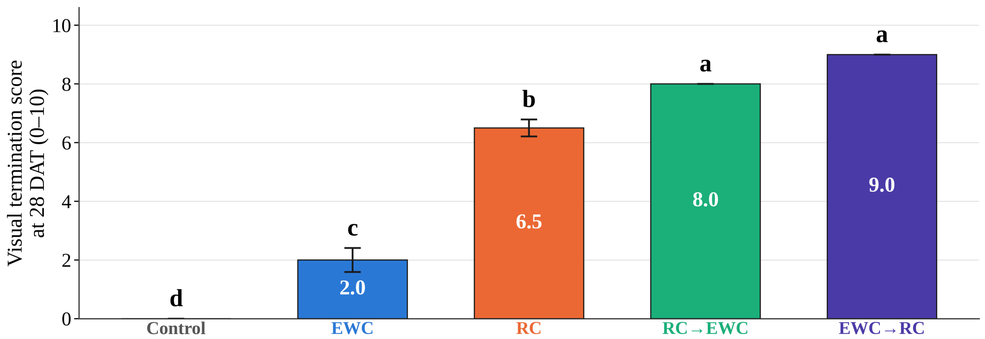


Fig 3. Visual termination score at 28 DAT (0 = none, 10 = complete; mean ± SE, n = 4); P < 0.001. Letters: Tukey's HSD (α = 0.05).

Key results

- **Termination:** at 28 DAT, EWC→RC (9.0) and RC→EWC (8.0) beat RC (6.5) and EWC (2.0) (Fig. 3); order did not differ (P = 0.08), consistent with crimping + flaming **(Frasconi et al. 2019)**
- **Regrowth:** 79% of the rye survived EWC alone and 34% survived RC, but only 6–11% survived the sequences (Fig. 5), consistent with EWC top-kill of rye **(Sullivan 2025)**
- **Biomass and soil water:** the sequences cut biomass by 57–62% (Fig. 6); crimped plots held 27–28% soil moisture vs 17% in the control, as with rolled rye mulch **(Kornecki et al. 2012)**



Fig 4. Proposed mechanism (conceptual, not measured): EWC cannot terminate rye completely and the crimper leaves some tillers standing; together, few tillers survive.

Conclusions

- **Electrocution improved roller-crimper termination:** one same-day EWC pass raised termination from 65% to 80–90% and cut surviving rye from 34% to 6–11%.
- **Results supported the hypothesis:** EWC + RC beat either tool alone, in either order. EWC alone could not terminate rye, even at ≈ 10× the dose that terminated an oat–clover mix **(WSU 2025)**, likely because mature rye (Zadoks 85) holds little tissue moisture and its lignified tissue conducts current poorly.

TAKEAWAY

Add a same-day EWC pass to roller crimping, in either order: termination rose to 80–90% and surviving rye fell to 6–11%.

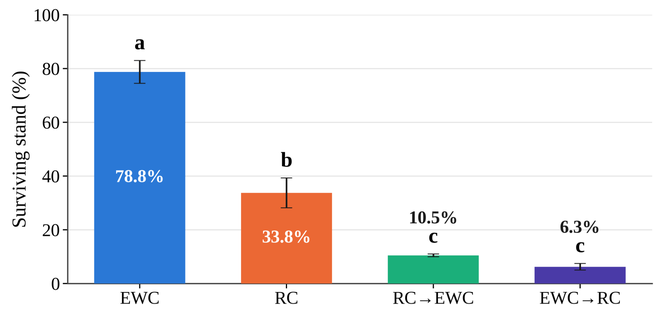


Fig 5. Surviving stand at 15 DAT (mean ± SE, n = 4); P < 0.001. Letters: Tukey's HSD (α = 0.05).

What it shows: share of rye plants still alive and regrowing 15 days after termination; lower means a more complete kill.

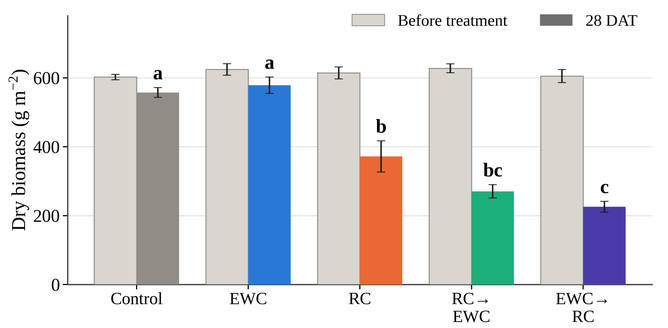


Fig 6. Dry biomass before and 28 DAT (mean ± SE); initial P = 0.75. Letters compare 28 DAT.

Why biomass fell: killed, flattened rye lies on moist soil where microbes decompose it, while surviving tillers keep adding biomass **(Peterson et al. 2025)**.

Future Research

- Test earlier termination (anthesis) and multiple EWC passes
- Evaluate other cover crops, weed suppression and cash-crop yield

References

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